

SESSION-06
SECURITIES ANALYSIS & PORTFOLIO MANAGEMENT

From Return Measurement to Portfolio Pricing

A Capital Market Theory Journey



SESSION 1

What did I earn?

Return & risk measurement



SESSION 2

How do assets interact?

Covariance, correlation, beta, alpha



SESSION 3

What's the best combination?

Portfolio risk-return & diversification



SESSION 4

Where's the efficient boundary?

Efficient frontier, MVP, CML



SESSION 5

What should I be paid for risk?

CAPM & SML



SESSION 6

Did my manager add value?

Risk-adjusted performance evaluation

"By the end of these 6 sessions, you should be able to open any mutual fund fact sheet and know exactly which of these six questions its marketing is quietly avoiding. Which one would you guess — and why?"

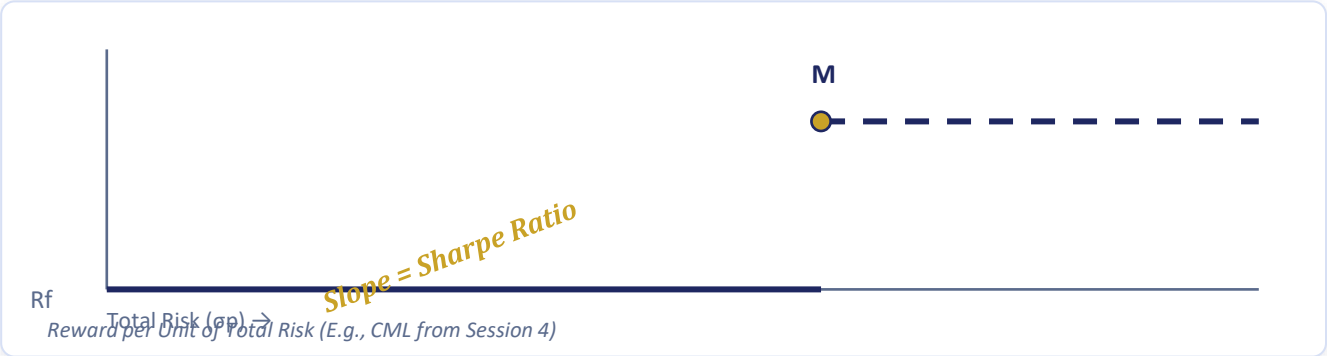


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The Sharpe Ratio: Reward per Unit of Total Risk

- Sharpe Ratio = $(R_p - R_f) / \sigma_p$
- This is literally the CML's slope, applied to any portfolio — using total risk as the denominator
- Appropriate for evaluating an investor's entire wealth (undiversified exposure) — it penalizes a manager for risk the investor could have diversified away elsewhere
- Two portfolios with the same return can have very different Sharpe ratios once correlation/total risk differs
- Reconnects directly to Session 3's Slide 22 example: Sharpe = 0.43 ($\rho=+0.8$) vs 0.66 ($\rho=-0.3$)



SESSION 3 · SLIDE 22 COMPARISON

Portfolio	Correlation (ρ)	Sharpe Ratio
Portfolio X	+0.8	0.43
Portfolio Y	-0.3	0.66

Both portfolios: $E(R_p) = 10\%$ — identical return, very different risk-adjusted outcome.

REFLECTION POINT

“If Sharpe ratio penalizes a manager for total risk, is that fair to a manager whose portfolio is only one small piece of a client’s much larger, already-diversified wealth?”

QUESTION BANK

None directly on this slide — feeds immediately into Problem 8.1 on the next slide.

Active Fund vs Passive Index: A Fee-Aware Evaluation

- Compute and compare Sharpe ratio, Treynor’s ratio, and Jensen’s alpha for two competing large-cap funds
- Active-vs-passive is not a single-metric decision — a fund can win on Sharpe but lose on Jensen’s alpha, or vice versa
- Fee drag must be applied before, not after, the risk-adjusted comparison
- Decision-relevant question: is the fee premium justified by risk-adjusted, not raw, outperformance?

Metric (Net of Fees)	Fund A — Active Large-Cap	Fund B — Passive Index
Expense Ratio	1.8% p.a.	0.2% p.a.
Sharpe Ratio	0.51	0.58
Treynor’s Ratio	9.8	9.5
Jensen’s Alpha	+0.30%	0.00%
Verdict	Wins on Jensen’s alpha — but the 1.6% fee premium is not fully justified once Sharpe is considered	

REFLECTION POINT

“An active fund shows a higher raw return than its benchmark index. After fees and full risk-adjustment, would you still recommend it to an endowment committee?”

QUESTION BANK

IN-CLASS Problem 8.1 (active large-cap fund versus index fund, endowment context)

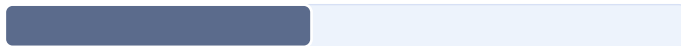
Skilled or Just Concentrated? Sharpe vs Treynor vs Jensen's Alpha

- Apply Sharpe, Treynor, and Jensen's alpha to a concentrated PMS portfolio
- The “right” ratio depends on what the portfolio represents for the investor: entire wealth → Sharpe (total risk); one holding within a diversified whole → Treynor / Jensen's alpha (systematic risk only)
- A concentrated manager can look excellent on Treynor (ignores unsystematic risk) while looking mediocre on Sharpe (penalizes the very concentration that drove the return)

• Judge whether outperformance reflects durable skill — or merely higher-risk taking

ONE CONCENTRATED PMS PORTFOLIO — THREE VERY DIFFERENT VERDICTS

Mediocre



SHARPE · TOTAL RISK

Excellent



TREYNOR · SYSTEMATIC RISK

Excellent



JENSEN'S ALPHA

Deliberately set at different “levels” to make the ranking-disagreement immediate — final ratio comparison only, no derivation shown.

REFLECTION POINT

“A PMS manager’s portfolio looks outstanding on Treynor but only average on Sharpe. If this is the client’s entire liquid net worth, which ratio should actually drive the renewal decision?”

QUESTION BANK

IN-CLASS Problem 8.2 (is the PMS manager skilled, or just concentrated?)

Choosing the Right Benchmark: Mid-Cap Fund Suitability for SIPs

- Compute Sharpe, Treynor, and Jensen's alpha for a mid-cap fund against its correct benchmark — not a large-cap index
- Benchmark selection error is a live, practical version of Roll's Critique (Session 5)
- What these ratios do not capture: liquidity risk, drawdown depth, and behavioural risk (an investor's ability to stay invested through volatility)
- Assess suitability for disciplined, long-horizon SIP investors — not just the headline ratio

BENCHMARK USED: NIFTY 50 (LARGE-CAP)

WRONG BENCHMARK

Jensen's Alpha = -1.2%

Jensen's Alpha turns negative — the fund looks like it destroys value

BENCHMARK USED: NIFTY MIDCAP 150

CORRECT BENCHMARK

Jensen's Alpha = $+2.1\%$

Jensen's Alpha turns positive — the fund genuinely adds value once measured against its true peer set

Same fund, same period — the sign of alpha flips purely on benchmark choice. Ratios also miss liquidity risk, drawdown depth, and investor behavioural risk.

REFLECTION POINT

"If using the wrong benchmark can flip a fund's measured alpha from positive to negative, how much should investors actually trust the 'alpha' figures quoted in fund marketing material?"

QUESTION BANK

IN-CLASS Problem 8.3 (should the mid-cap fund be included in SIP recommendations?)

Ranking Reversals: When Performance Measures Disagree

- Two pension fund managers with identical returns but different risk profiles
- Sharpe, Treynor, and Jensen's alpha can rank them in completely reversed order
- A ranking reversal is not a calculation error — it reflects genuinely different risk assumptions embedded in each ratio
- For a pension fund's liability-matching mandate: is total risk (Sharpe) or systematic risk (Treynor/Jensen) the right lens?
- Choosing the wrong ratio for the context can lead to the wrong manager being hired or fired

RANKED BY SHARPE (TOTAL RISK)

#1 Manager Alpha
Sharpe = 0.61

#2 Manager Beta
Sharpe = 0.44

RANKED BY TREYNOR / JENSEN'S ALPHA

#1 Manager Beta
Treynor = 11.2 | Jensen = +1.8%

#2 Manager Alpha
Treynor = 8.4 | Jensen = +0.6%

The rankings fully invert — Manager Alpha wins on Sharpe, Manager Beta wins on Treynor and Jensen's alpha.

REFLECTION POINT

"Given a pension fund's liability-matching mandate and its typical degree of diversification across multiple external managers, which single ratio should the trustees rely on most — and would your answer change for a different type of institutional investor?"

QUESTION BANK

IN-CLASS Problem 8.4 (two pension managers, three conflicting signals)

Capital Allocation Across Sleeves: The Family Office Decision

- Compute Sharpe, Treynor, and Jensen’s alpha for three portfolio sleeves
- Allocate the family office’s next rupee by weighing these ratios against governance constraints: benchmark validity, liquidity, fees, and mandate
- The course’s most integrative numerical problem — forces synthesis of every performance-evaluation concept into a single real-world capital-allocation decision

Metric	Sleeve 1: Domestic Large-Cap	Sleeve 2: Global Equity Feeder	Sleeve 3: PMS / Alternatives
Sharpe Ratio	0.58	0.71	0.66
Treynor’s Ratio	9.6	12.1	10.8
Jensen’s Alpha	+0.4%	+1.6%	+1.1%
Liquidity	High	Medium	Low (lock-in)
Benchmark Validity	Strong	Weak	Strong
Fee Level	Low	High	Medium

Allocation call: Sleeve 2 posts the highest raw ratios, but weak benchmark validity and high fees temper conviction — the next rupee is split between Sleeve 1 (liquidity anchor, strong benchmark) and Sleeve 3 (differentiated, genuine alpha), not simply chasing Sleeve 2’s headline numbers.

REFLECTION POINT

“If the highest-Sharpe sleeve is also the least liquid and has the weakest benchmark validity, should it still receive the family office’s next rupee? What would you tell the principal?”

QUESTION BANK

IN-CLASS Problem 8.5 (family office: where does the next rupee go?) — recommended as the capstone class problem.

What Institutional Investors Actually Do: Theory vs Practice

- The gap between theory and practice is not a failure of theory — it is a natural evolution built on top of the theory
- Grounds the entire course's apparatus in what students will actually encounter in asset management, private banking, or corporate treasury roles

TEXTBOOK THEORY

- Mean-variance optimization
- CAPM-based required returns
- Efficient frontier construction
- Beta as the sole risk metric
- Homogeneous expectations
- Single-period optimization
- No transaction costs
- Unlimited borrowing at R_f

ACTUAL PRACTICE

- Multi-factor risk models (Barra, Axioma)
- Risk budgeting & factor attribution
- Smart beta / factor tilts (value, low-vol)
- Liability-driven investing (LDI)
- Tail-risk hedging & scenario analysis
- Black-Litterman (views + prior)
- ESG integration & engagement
- Alternatives (PE, infrastructure, real assets)

REFLECTION POINT

"If institutional practice has moved this far beyond textbook CAPM and Markowitz, why do we still start every finance education with these exact models?"

QUESTION BANK

None (industry-context slide).

The Indian Market Perspective: Household Allocation & Retail Behaviour

- Over 70% of household wealth sits in physical, illiquid assets — a classic home-bias / illiquidity pattern
- Mutual fund AUM has crossed ₹58 lakh crore, yet equity penetration remains below 5% of households — a gap between industry growth and household diversification
- NSE & BSE host 5,000+ listed securities; Nifty 50/Sensex are domestic proxies that exclude real estate, gold, and informal-sector assets — a live example of Roll's Critique

Indian Household Asset Allocation (approx.)



POST-COVID RETAIL SURGE

100M+

demat accounts opened since the pandemic

Concentration risk: growth is skewed toward F&O trading and high-beta small caps — unsystematic, uncompensated risk at genuine scale.

REFLECTION POINT

“Given everything this course has covered — diversification, correlation, systematic vs unsystematic risk — what specifically is wrong, from a portfolio-theory standpoint, with a household holding 70%+ of its wealth in real estate and gold?”

QUESTION BANK

None (India-context grounding slide).

The Great CAPM Debate: Should It Still Be Taught?

Revisits the course's central unresolved tension, first raised in Session 2 and Session 5.

MOTION: *"CAPM should no longer be taught as the primary asset-pricing model."*

FOR THE MOTION

- Empirical evidence consistently rejects the linear beta-return relationship
- Low-beta anomaly, size, value, and momentum cannot be explained by beta alone
- Roll's critique makes CAPM empirically untestable by construction
- Multi-factor models (Fama-French-Carhart) demonstrably outperform CAPM
- Behavioural biases structurally violate the homogeneous-expectations assumption

AGAINST THE MOTION

- CAPM provides an elegant, intuitive foundation for risk-return education
- No single agreed replacement exists — APT, FF3, FF5 all face their own limitations
- CAPM is still used by the majority of CFOs for capital budgeting (Graham & Harvey survey evidence)
- The equilibrium logic remains valid: systematic risk demands higher return
- Teaching any alternative still requires CAPM as the conceptual baseline

REFLECTION POINT

"Assign half the room to argue 'for' and half to argue 'against' this statement — using only evidence already covered in this course. After both sides present, would you personally still recommend CAPM be taught as the primary model?"

QUESTION BANK

None directly (capstone debate) — draws on the cumulative evidentiary base of Problems 4.5–4.9, 7.1–7.5, and 8.1–8.5.

Retain, Modify, Question: Closing Synthesis and Discussion

● RETAIN

- Diversification as a non-negotiable discipline
- The risk-return trade-off framework
- Asset allocation as the primary driver of outcomes
- Unsystematic risk earns no premium (Session 2, Slide 14)

● MODIFY

- Beta-only risk measurement → multi-factor models
- Single-period optimization → dynamic asset allocation
- Frictionless markets → transaction-cost constraints
- Homogeneous expectations → scenario analysis & stress tests

● QUESTION

- Normality of returns — fat tails exist
- Static correlations — they spike in crises (Session 3, Slide 26)
- Market portfolio observability (Roll's critique)
- The rational-investor assumption (behavioural finance evidence)

REFLECTION POINT

"If managing ₹1,000 crore today, which principles from Markowitz and CAPM would still guide your decisions? What would you discard?"

QUESTION BANK

None (final synthesis — implicitly draws on the entire course).